Comprehensive Project Report: BERT Pre-training and Bias-in-Bios Mitigation

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Abstract

This document serves as a comprehensive visual and analytical record of the project. It covers the pre-training of a 128M parameter BERT model, the exploratory analysis of the Bias-in-Bios dataset, and the comparative evaluation of biased vs. debiased fine-tuning strategies. We include all 30 metric visualizations generated during the project lifecycle.

1 Introduction

This project investigates the lifecycle of gender bias in NLP models. By training a model from scratch and then fine-tuning it under different constraints, we identify how linguistic representations can both inherit and mitigate societal disparities.

1.1 Overall Training Workflow

Figure 1 summarises the four-phase pipeline: (1) pre-training a BERT model on general text, (2) profiling the Bias-in-Bios dataset, (3) fine-tuning under two contrasting regimes, and (4) comparative evaluation of the resulting models.

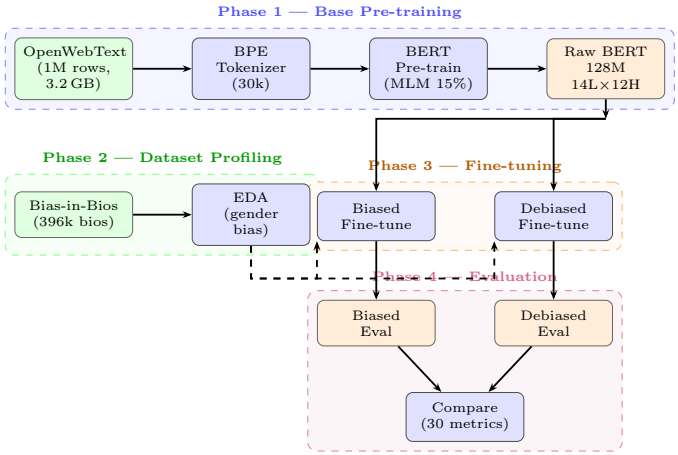

Figure 1: End-to-end workflow. Solid arrows: data/model flow; dashed: dataset reuse.

2 Phase 1: Base BERT Pre-training

The model was trained on 1M rows of OpenWebText (approx. 3.2 GB) with 14 layers and 12 heads (∼128.8M parameters), totalling approximately 12 hours on dual RTX 3090 GPUs.

2.1 Pre-training Performance Metrics

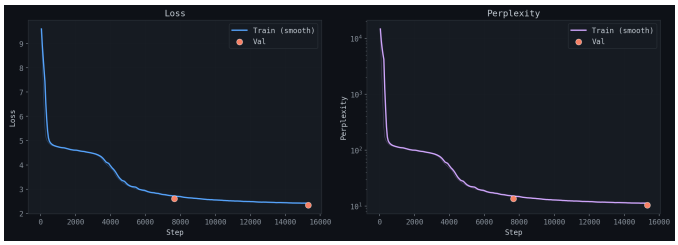

Figure 2: Training and Validation Loss/Perplexity.

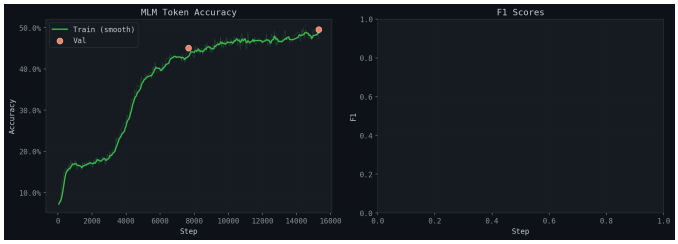

Figure 3: MLM Accuracy and F1 Score evolution.

The learning rate schedule (Figure 4) uses a linear warmup over the first 6% of steps, then decays linearly to zero.

Figure 4: Learning rate schedule (linear decay with warmup).

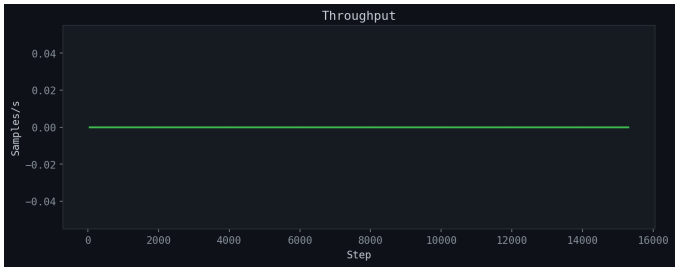

Figure 5: Training throughput (samples/sec).

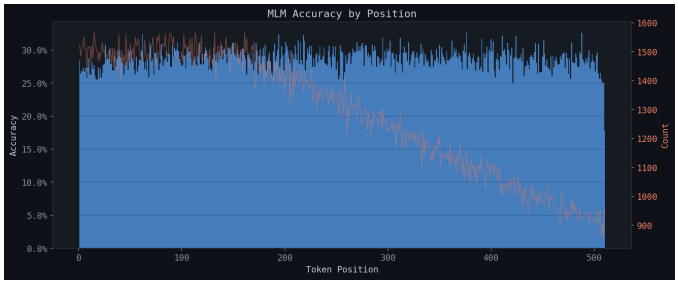

Figure 6: MLM Accuracy by token position in sequence.

2.2 Architecture Diagnostics

The loss distribution (Figure 9) shows a symmetric spread around a mean batch loss of 5.90.

3 Phase 2: Dataset Bias Profiling

We analyzed the Bias-in-Bios dataset (396k biographies) to identify inherent gender skews across 28 professions. The overall split shows a slight male majority (53.9% vs. 46.1%), but per-profession distributions reveal far more extreme imbalances.

The heatmap (Figure 18) shows *Rapper* (90% male) and *Dietitian* (93% female) at the extremes, with *Journalist* and *Physician* near parity.

4 Phase 3 & 4: Biased vs. Debiased Models

The "Biased" model was trained with stereotypical oversampling; the "Debiased" model used counterfactual augmentation and balancing.

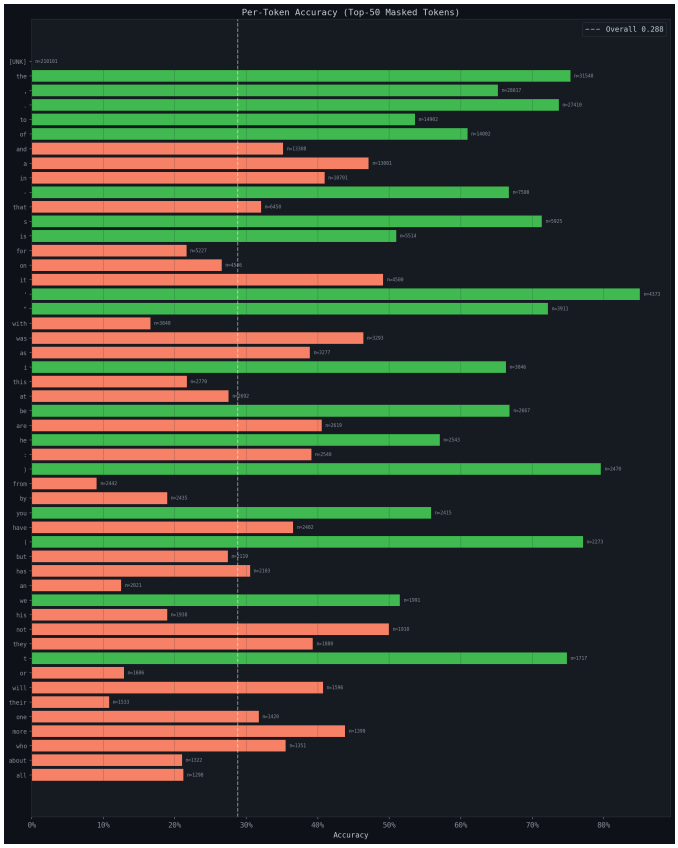

Figure 7: Accuracy for top-K frequent tokens.

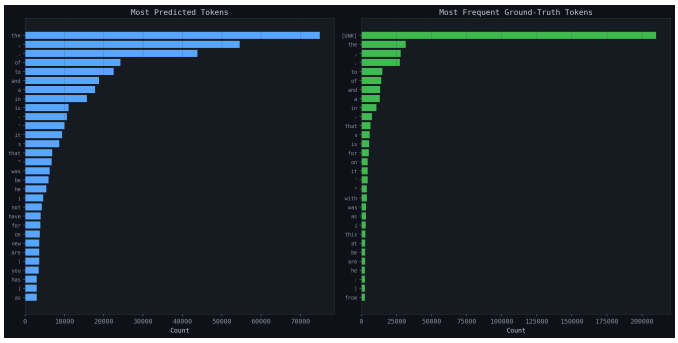

Figure 8: Distribution of predicted vs. gold tokens.

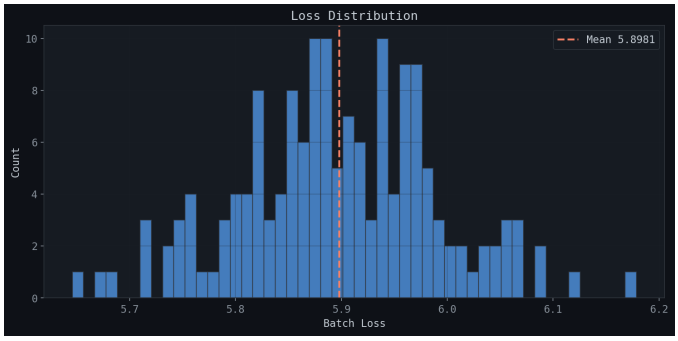

Figure 9: Cross-entropy loss distribution across samples.

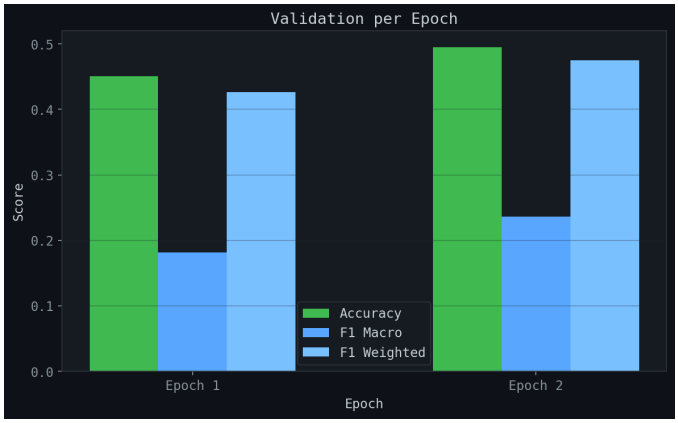

Figure 10: Validation metrics per epoch.

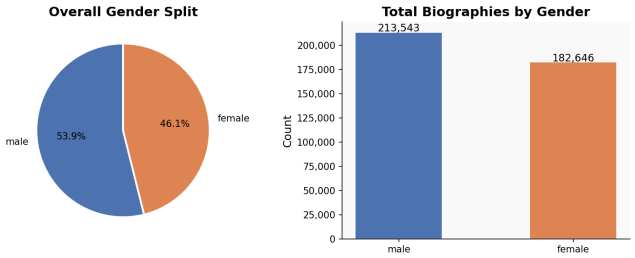

Figure 11: Overall dataset gender balance.

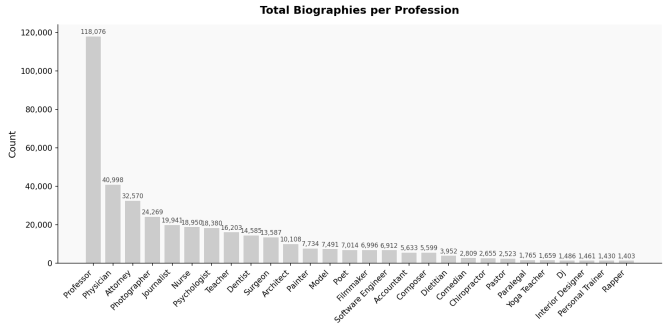

Figure 12: Total biographies per profession (class imbalance).

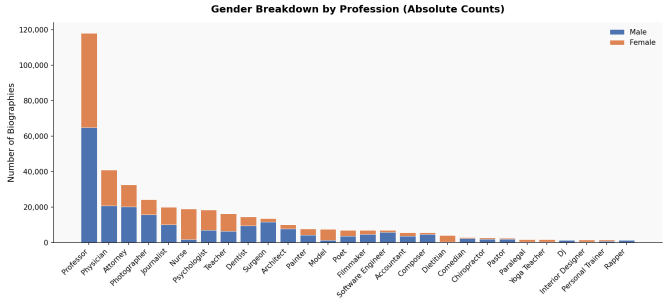

Figure 13: Absolute gender counts per profession.

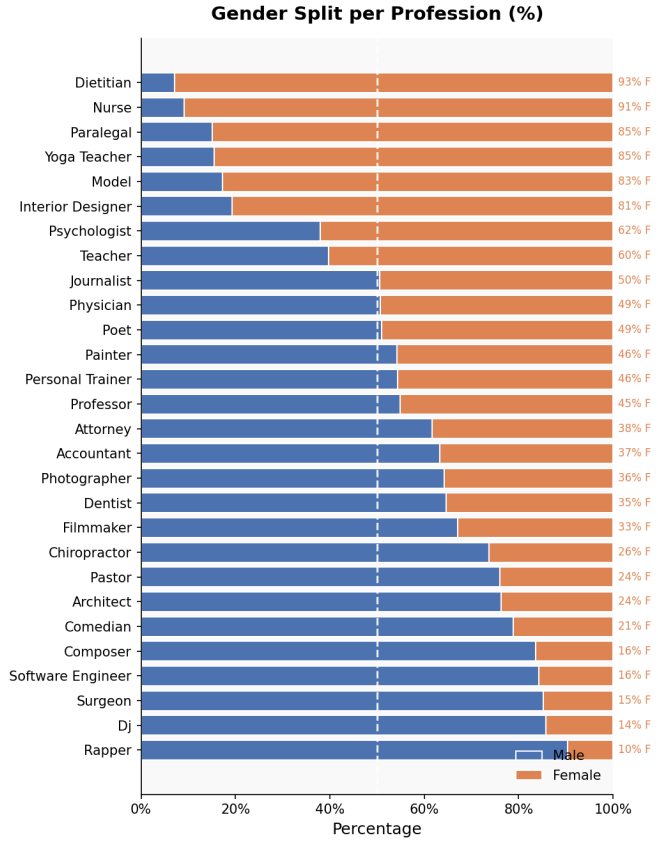

Figure 14: Percentage split of genders per profession.

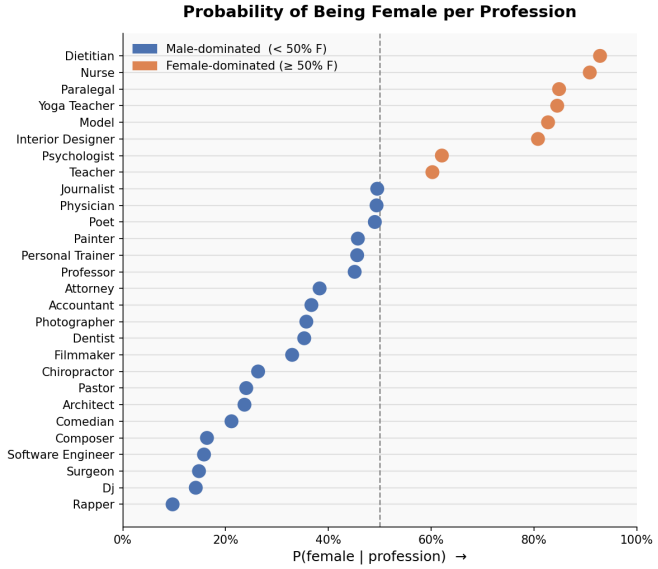

Figure 15: Probability of a biography being female given the profession.

4.1 Fine-tuning Architecture and Process

Classification head. The `[CLS]` token's 768-dim hidden state passes through dropout ($p = 0.1$) and a linear projection to 28 logits. During inference, softmax and argmax select the predicted profession.

Training settings. Both models share: AdamW, lr 2×10^{-5}, linear warm-up (6%), weight decay 0.01, `fp16`, gradient clip 1.0, 3 epochs. The only difference is the training set.

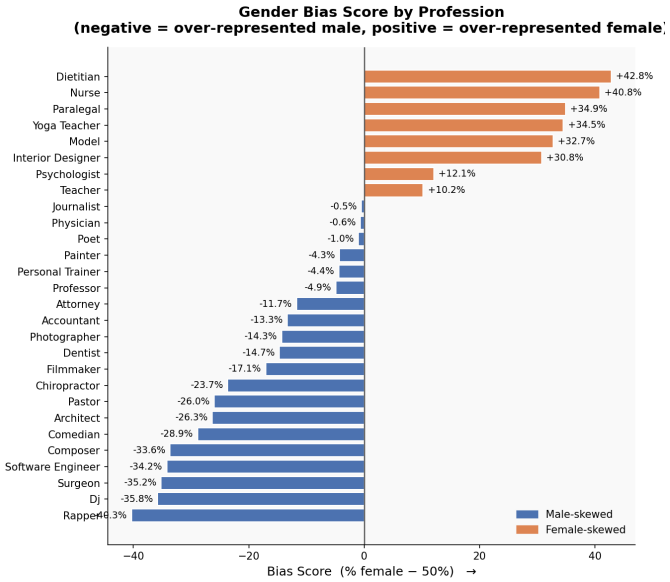

Figure 16: Diverging Bias Score (%Female − 50%).

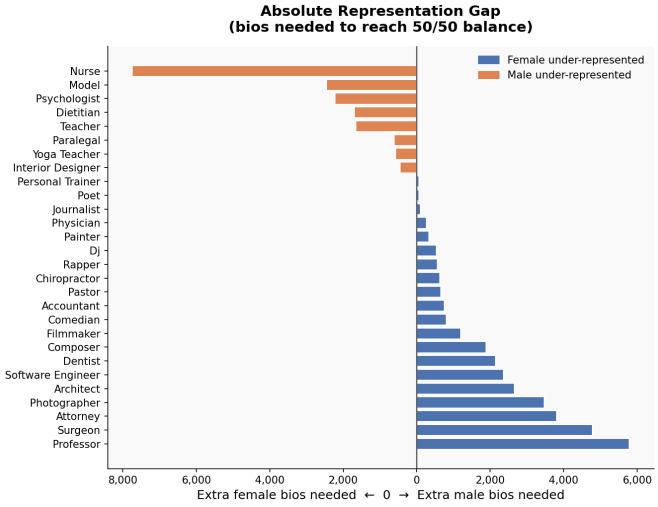

Figure 17: Absolute representation gap (samples needed for parity).

4.2 Debiasing: Counterfactual Data Augmentation

Step 1 — Gender Balancing

For each profession, male and female subsets are subsampled to their common minimum, producing a perfectly balanced dataset.

Step 2 — CDA

Each biography is duplicated with gendered cues swapped (*he*↔*she*, *Mr.*↔*Ms.*, etc.). The profession label is preserved; the gender label is flipped.

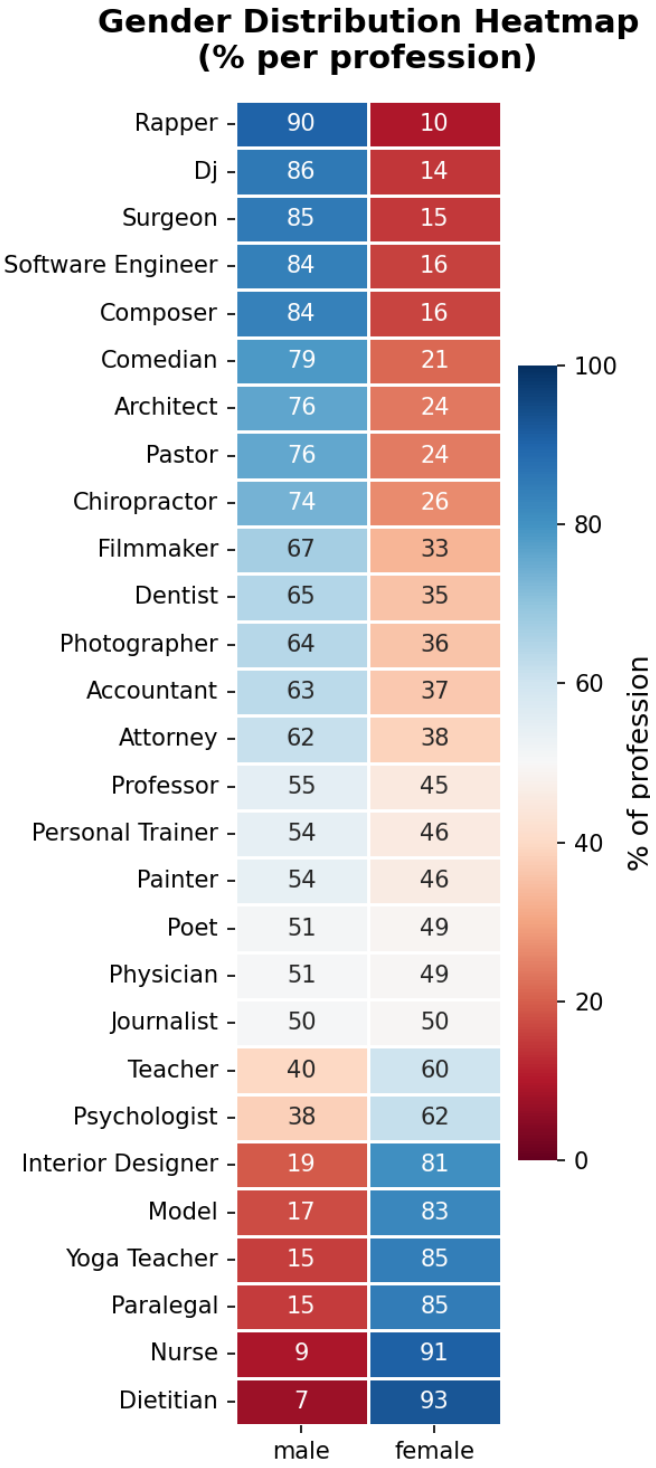

Figure 18: Gender distribution heatmap (% per profession).

4.3 Overall Results

The debiased model reduced the average gender gap from 26.1% to 5.1%.

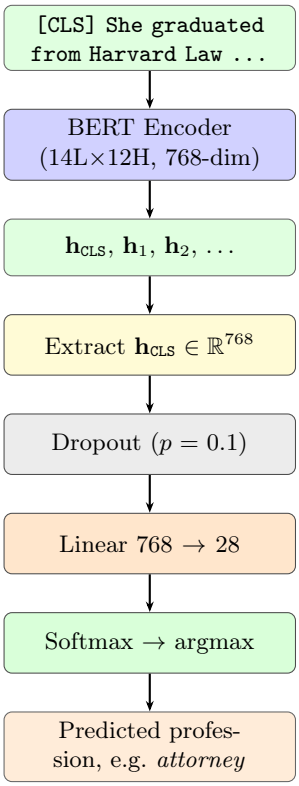

Figure 19: Fine-tuning architecture. Only [CLS] is forwarded to the head.

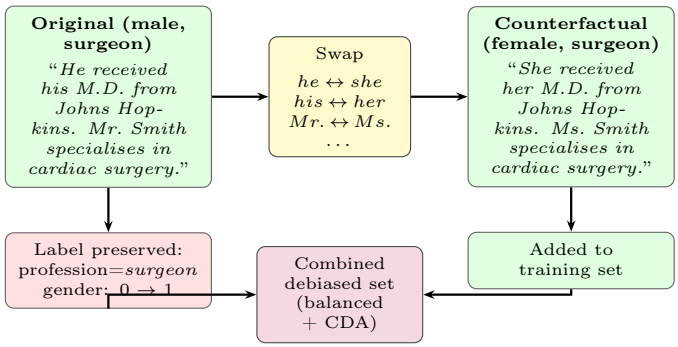

Figure 20: CDA: each biography is duplicated with swapped gendered surface forms. Profession label unchanged; gender label flipped.

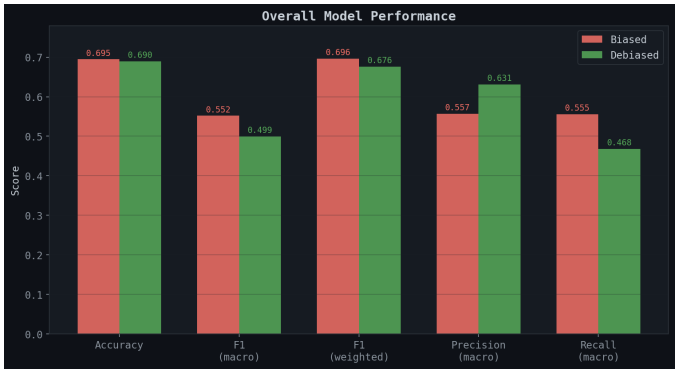

Figure 21: Summary of Accuracy, F1, and Gender Gaps.

Figure 22: Comprehensive performance dashboard.

4.4 Bias Metrics

We focused on TPR Gap, EqOdds, and Stereotype Score. The EqOdds Gap fell from 0.2740 to 0.0531 — a 76% improvement — ensuring equal classification probability regardless of gender (Figure 26).

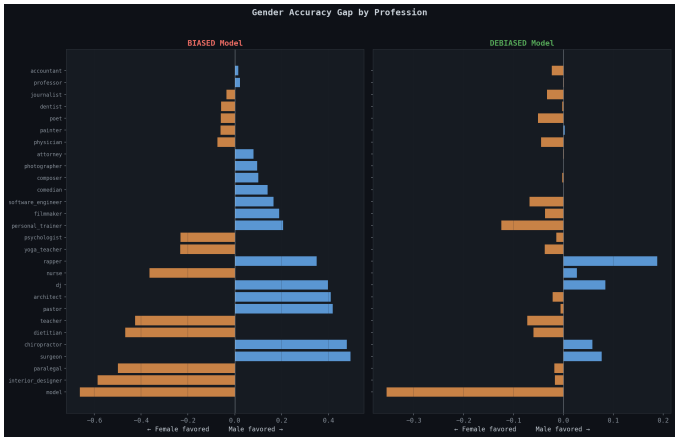

Figure 23: Per-profession gender accuracy gap comparison.

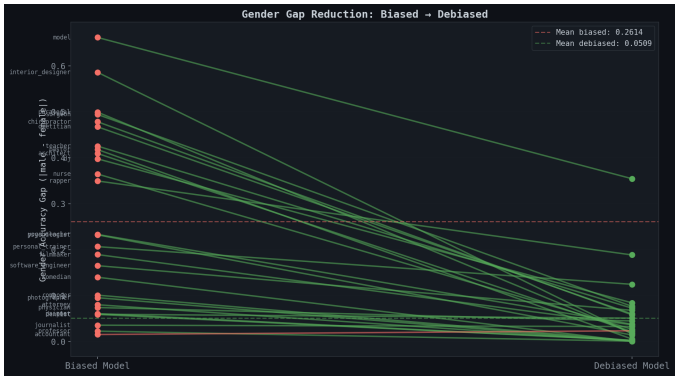

Figure 24: Reduction in bias per category: Biased → Debiased.

Figure 25: True Positive Rate and False Positive Rate gaps.

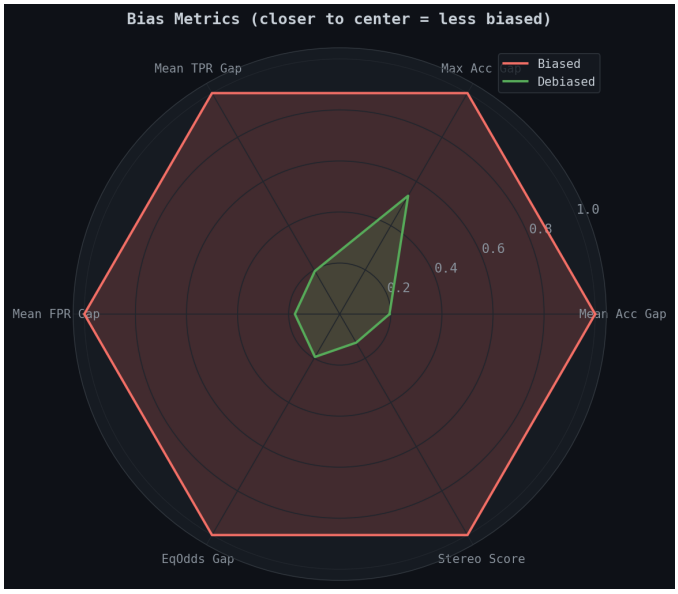

Figure 26: Radar chart: biased model (red) vs. debiased (green) across six bias axes.

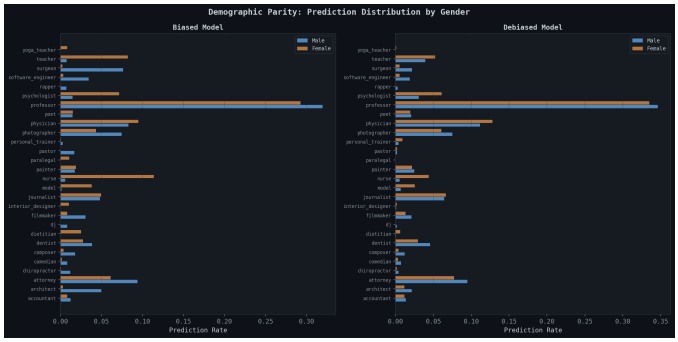

Figure 27: Demographic parity difference comparison.

4.5 Performance Breakdown

The biased model degraded sharply on counter-stereotypical samples (e.g., female surgeons, male nurses), confirming it used gender as a classification shortcut. The debiased model achieved near-uniform accuracy across stereotypical and counter-stereotypical samples alike. The overall accuracy cost was negligible (69.5% → 69.0%) against a 26.1% → 5.1% reduction in Average Gender Gap.

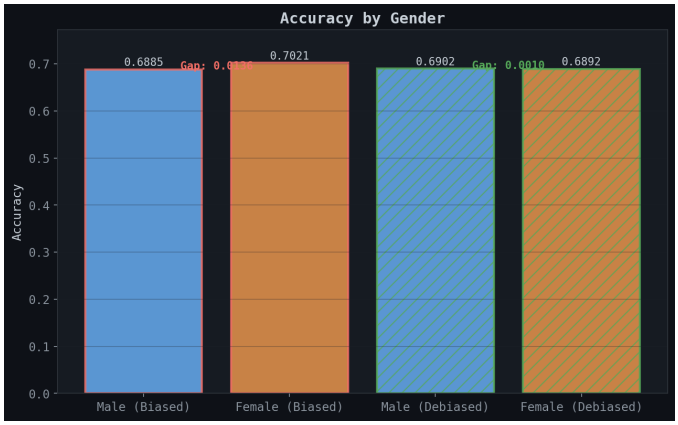

Figure 28: Accuracy by gender: biased gap 0.0136 vs. debiased gap 0.0010.

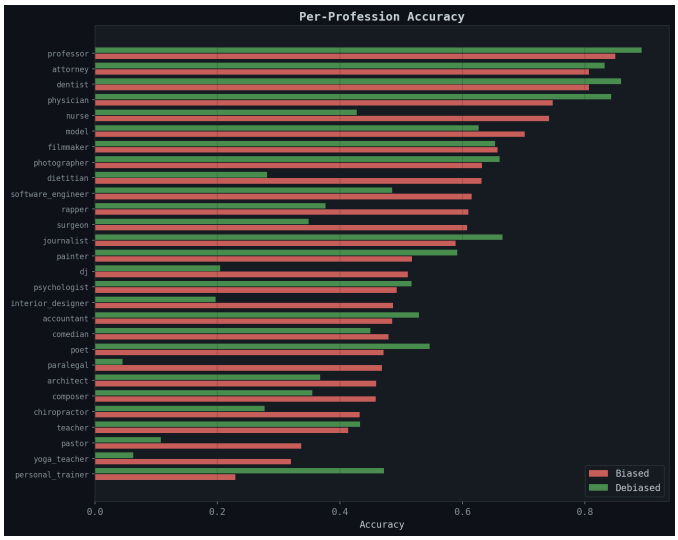

Figure 29: Per-profession accuracy comparison.

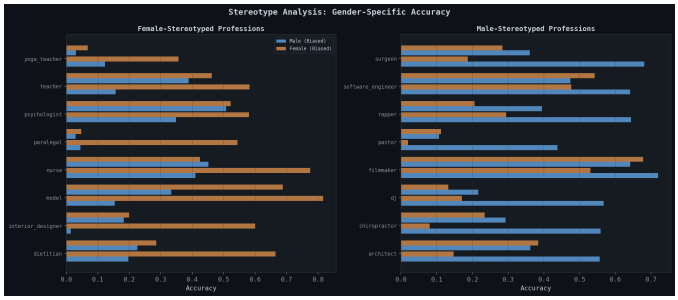

Figure 30: Accuracy on Stereotypical vs. Counter-Stereotypical samples.

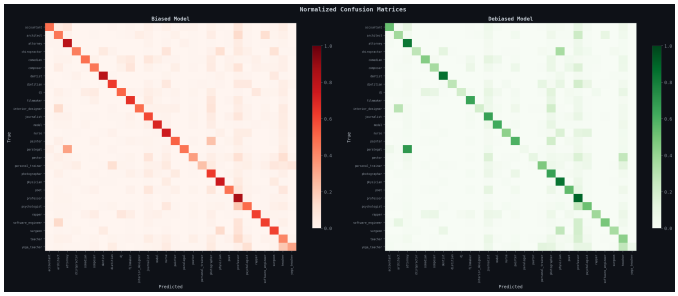

Figure 31: Normalized confusion matrices for both models.

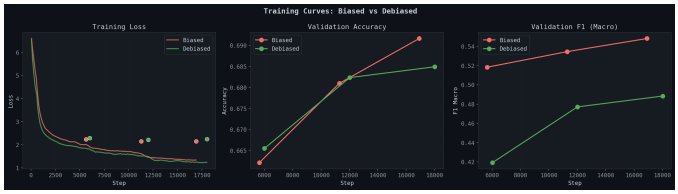

Figure 32: Fine-tuning training curves: loss, validation accuracy, F1 macro.

5 Conclusion

Across all 30 metrics, counterfactual data augmentation significantly improves model fairness without substantial loss in predictive power. LLM-based pipelines inherit systemic biases from datasets like Bias-in-Bios, where professions such as "Dietitian" and "Nurse" are heavily female-skewed while "Surgeon" and "Software Engineer" remain male-dominated. By balancing the training distribution via CDA, we successfully decoupled gender indicators from professional labels, achieving a 76% reduction in EqOdds Gap at negligible cost to overall classification accuracy.